[Lecture Note] Results and Discussion

GE2021 Research Technology, Spring 2025

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Sun, 9 March 2025

Learning Objectives

- 13-1 Write a results section.
- 13-2 Describe how to present results in quantitative studies.
- 13-3 Explain how to present results in qualitative studies.
- 13-4 Prepare visual representations of results.
- 13-5 Compose a discussion section.
- 13-6 Develop a recommendations section based on methodology or topic.

SHORT VIDEO INTRODUCTION

https://www.youtube.com/watch?v=6J4m0gLddog

Reporting Results

- The best way to write is by starting with the section that feels most comfortable. Then, the researcher can expand, think, analyze, converse, and edit as well as add to the other sections as the work progresses.
- It is helpful to think about organization before diving in.

Demographic Section

- The results section for both qualitative and quantitative designs generally begins with the demographics of the sample.
- Basic characteristics of the population, such as gender, race and ethnicity, age, height, weight, marital status, sexual orientation, religious affiliation, employment status, and income are included, as well as the number of participants in the study.

- The demographic information is important, even if the focus of research is somewhere else, seemingly unrelated to demographics.

Findings Section

- After reporting demographics, the results section reports the findings of the study.
- In quantitative studies, researchers typically provide general information about the variables using measures such as frequency, averages, or percentages to give a broader picture of what the researcher found to be of interest.
- In qualitative studies, the results will list and discuss the codes, patterns, and themes found in the study.

Results in Quantitative Studies

- A straightforward statement of findings is customary, written in a direct and matter-of-fact style.
- Tables and figures are strongly recommended because they allow a visual presentation of detailed data.

Rejecting the Null Hypothesis

- Being able to reject the null hypothesis may not mean that we have proven the alternative hypothesis.
  - There are often scenarios in which we reject both the null hypothesis and the alternative hypothesis, because the relationship between variables is different than predicted.

Failing to Reject the Null Hypothesis

- Failing to reject the null hypothesis still contributes to the advancement of science, because it means that the null hypothesis stands as an explanation for the relationship between variables of interest.
- When we fail to reject the null, we are sometimes able to identify new relationships between variables or discover other unexpected associations.

Results in Qualitative Studies

- Results in qualitative research follow a similar format to those in quantitative studies, providing an overview of demographics and then pointing out patterns from the analysis.
- The most straightforward way to report results is to rank them in order of importance and then combine results where patterns exist.
- To support their point, researchers use quotations directly from the text, illustrate the findings with excerpts from the data, and write a complex narrative as their results.

Visually Presenting Results

- Data visualization: visually telling story through data; a growing discipline in its own right that cuts across all other fields.
- They bring meaning, show dynamics, depict details, and ultimately illuminate the key relationships researchers and readers should be focusing on.
- The most widely used types of visual presentations are tables and figures.

Tables

- Tables can deliver a large amount of information at one time.
- They work best when the information is extensive and providing it in narrative form can be inefficient.
- A good rule of thumb is to present the information in a table when it occupies less space than a written rendition of the same information.
- Refrain from repeating the outcomes in a table and in the narrative; discuss the results and their implications, but do not repeat the points that have been made visually.

Figures

- Some common types of figures are charts, graphs, plots, and path diagrams.
- Path diagrams: a basic model that depicts the relationship between variables by using arrows or other types of connectors.
- It is important to consider the type of figure to use and its purpose.
- It is then helpful to consider other stylistic guidelines, such as labels, colors, amount of information, and clarity.
- Having a good numbering or labeling system will keep figures in order and ensure the reader can reference them.

Discussion

- The discussion is where you write about your interpretation of the results and what comes next.
- The discussion should begin with a clear outline of what will be included and how, such as delineating the limitations of the study.
- Limitations explore the factors that could have improved the effectiveness, such as a bigger sample for more reliable results or a more diverse sample.
- One way of organizing the discussion of findings would be by order of importance.
- The interpretation of results is the researcher's opinion of the findings, but it is most often based on the literature or the theories previously researched.

Recommendations

- Toward the end of the discussion, most researchers anticipate how the study might be followed up with other studies to reveal even more knowledge and insight.
- Two common types of recommendations:
  - Recommendations on the methodology of future studies.
  - Recommendations on topics.

Class Activities:

- Select a peer-reviewed article from a database like PubMed and distribute to students to read in class, with the discussion section removed. In small groups, ask students to review the paper and identify the most salient findings they would choose to highlight in the discussion section of a peer-reviewed journal article.

- In this exercise, students simulate sending a manuscript they have produced, based on their research to an academic peer-reviewed journal. The journal reviewers respond to the students and point out several key limitations as listed in the table below. In small groups, ask students to discuss and complete the table below. The discussion should focus on:
  - Why the reviewer raised this as a limitation?
  - How you would respond to this limitation? Students can answer this question in a few ways, by (a) discussing how they might address this limitation in future research, (b) by downplaying this as a limitation, or (c) by portraying the limitation as a strength.

Study limitation	Why is this a limitation?	How would you respond to this limitation?
The number of individuals included in your study is too small.		
The data is self-reported.		
This is a cross-sectional design.		

- Identify and copy several tables and figures from various peer-reviewed research articles and distribute to students. Divide students into small groups and ask them to summarize, in words, takeaway from each table and figure.

House Keeping

Announcement

Assignment

Assignment1: Assignment Title

- **What to do**
  - A
  - B
- **Format and Requirement**
  - PDF format, < ?? pages
  - file name should be include your student id and name
    - stuID_name_title.pdf (e.g. 1111111_ChungilChae_SelfIntroduction.pdf)
  - APA, Double-Space, No title page, 12 points, Reference section if necessary
  - Chinese name in English, English name, Student ID, Class name and section (e.g. GE2021, W09; MGS3001, W01)
- due date
  - by Date(DAY) 11:59PM
  - NO LATE SUBMISSION ALLOWED!!!!

Reference